Austin Lin

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Open to Relocation

TECHNICAL SKILLS

Languages: Python, Java
Tools: MATLAB/Simulink, ROS2/Gazebo

EXPERIENCE

General Dynamics Electric Boat
Electrical Systems Engineer, New London, CT — Sept 2024 - Present

- Implemented a new shock modeling method for electrical cables, reducing excess cable length in a space-constrained environment by up to 60% compared to existing methods. Used MATLAB to implement statistical methods and cost-constraint optimization to fuse data from specifications, finite element shock studies, and other sources.
- Led a team of engineers to implement class-wide design changes to improve safety and reliability.
- Overcame production bottlenecks by providing engineering assistance to other disciplines (Design, Work Planning, Deckplate/Trades), supporting ongoing schedule acceleration efforts.

L3Harris C5 Systems
Systems Engineering Intern, Camden, NJ — June 2023 - Aug 2023

- Demonstrated product quality to the customer by performing System Operability Verification Tests for Marcom IVCS and Symphony. Completed CompTIA Network+ Training.
- Performed root-cause troubleshooting on network/radio configurations with Linux-based hardware.

UMD Space Systems Lab
Undergraduate Researcher, College Park, MD — May 2022 - Aug 2022

- Delivered spacecraft robot arm testing and modeling results to facilitate In-Space Servicing mission requirements. Conducted research with Benjamin Reed from Quantum Space and Dr. David Akin.
- Calculated and visualized the workspace of a robotic arm to reach payloads for in-orbit servicing. Used Denavit-Hartenberg parameters to calculate forward kinematics.
- Designed and fabricated a ¼-scale physical robotic arm prototype. Produced and executed a test procedure to validate Python model results for 24+ spacecraft configurations.

PROJECTS

Battery Simulation and State of Charge Estimator

- Created a Python/Shiny app to visualize the impact of adjusting Luenberger observer gain and measurement noise on battery state of charge estimation accuracy and convergence.

Robotics and Autonomous Systems

- Implemented a maze-solving algorithm on a robot using LIDAR for live localization and mapping. Used Gazebo and ROS2 to model a virtual environment to tune the algorithm before field testing.
- Programmed robots to identify, grab, and hand off LEGO bricks to each other using computer vision. Trained a YOLOv8 model for object identification and odometry, and used ZeroMQ through TCP/IP for communication.
- Developed hardware and software for a robot arm and controller. Programmed C++ firmware for an Arduino microprocessor to handle serial communication to a computer that calculates and graphically displays arm kinematics.

EDUCATION

University of Maryland, College Park, MD — Aug 2020 - Aug 2024
B.S. Mechanical Engineering, Minor in Robotics and Autonomous Systems
Scholars Public Leadership Program